MIL-S-21859E
4 February 1985
SUPERSEDING
MIL-S-21859D
29 July 1975

MILITARY SPECIFICATION

SUPPORT SETS, STACKING, MATERIAL HANDLING PALLET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 8000-pound capacity knockdown support sets for stacking flush type, two-way entry or wing type, four-way entry materials handling pallets.

1.2 Classification. The support sets shall be of the following sizes, as specified (see 6.2):

Size 1 - 38 inches long by 49-1/16 inches wide, (40 by 48 pallet).
Size 2 - 46 inches long by 61-1/16 inches wide, (48 by 60 pallet).
Size 3 - 30 inches long by 41-1/16 inches wide, (32 by 40 pallet).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-N-105 - Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3990

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NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13216E9021	- Support Set, Stacking, Material Handling, Pallet, Size 1.
TA13216E9026	- Support Set, Stacking, Material Handling, Pallet, Size 2.
TA13216E9027	- Support Set, Stacking, Material Handling, Pallet, Size 3.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pallet support sets (hereinafter called "support sets") shall be in accordance with the applicable drawings, and as specified herein. The support sets, assembled to simulate a single warehouse stack, shall support a 10,000 pound proof load and a side force of 200 pounds without any evidence of permanent deformation exceeding 1/16 inch, damage, or failure (see 4.5.2.1).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the support set. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inpsection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), within the time frame specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of support sets from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish support sets conforming to this specification.

3.3 Materials. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the support sets may be newly fabricated from recovered materials to the maximum extent practicable, provided the support sets produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the support sets.

3.4 Treatment and painting. The support sets shall be treated and painted in accordance with MIL-T-704, type A, or equivalent. The finish coat color shall match FED-STD-595, color No. 13538.

3.5 Identification marking. The support sets shall be identified in accordance with MIL-STD-130.

3.6 Support set size 1. Size 1 support sets shall be as shown on TA13216E9021 and as specified herein.

3.7 Support set size 2. Size 2 support sets shall be as shown on TA13216E9026 and as specified herein.

3.8 Support set size 3. Size 3 support sets shall be as shown on TA13216E9027 and as specified herein.

3.9 Workmanship. All parts and assemblies shall be clean and free from fins, sprues, scale, flux, and other extraneous material. Burrs shall be removed and sharp edges shall be rounded. No parts shall be damaged or impaired in any way, and shall show no evidence of corrosion.

3.9.1 Welders and welding.

3.9.1.1 Welder qualification. Welder qualification shall be in accordance with the American Welding Society Structural Welding Code, D1.1, Section 5, Qualification.

3.9.1.2 Welding. Welding of steel shall be in accordance with the American Welding Society procedures for the applicable steels, welding process optional. When welding is done after the item has been painted, the welded area shall be recleaned, treated, and painted in accordance with 3.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsbile for assuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The support set shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The support set shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. The inspection level shall be level II with an AQL of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. The inspection level shall be S-3 with an AQL of 1.0 percent defective.

4.5 Inspection procedure

4.5.1 Examination. Support sets shall be examined as specified herein, for the following defects:

Major

101. Support sets incomplete.
102. Any design characteristic not in accordance with the specified requirements.
103. Any component missing or not specified type.
104. Weld missing where required.

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105. Weld not specified type or size.
106. Weld not continuous or weld fractured, undercut, burned through, porous, or incompletely fused.
107. Metal deformed in welding.
108. Rust, scale, or other foreign matter at weld.
109. Tubing bent, cracked, crushed, fractured, or malformed.
110. Angles bent, broken, or malformed.
111. Dimension not as specified on the drawings.

Minor

201. Not finished or finish color not as specified.
202. Finish not adherent, i.e., peeling, blistered, or flaking.
203. No film (prime coat showing through).
204. Sharp edge, burr, or sliver not as specified.
205. Identification marking missing, illegible, or incorrect.

4.5.2 Test.

4.5.2.1 Proof load test. Assemble the support sets and erect a simulated single warehouse stack five pallets high. Place a uniformly distributed load of 2500 pounds on each of the four support pallets. (A similar load may be placed on the bottom pallet.) Allow the simulated stack to remain for not less than 5 minutes. Apply a side force of 200 pounds, perpendicular to the side of the stack, to the top pallet for not less than 90 seconds. Observe the stability of the stack. Repeat the side loading, applying the force to each of the remaining three sides of the top pallet. Disassemble the stack and examine the support set components. Loss of stack stability, or permanent deformation exceeding 1/16 inch, damage to, or failure of any component shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.1.1 Unit of product. For the purpose of inspection, a unit of product shall be a completed pack of any one size of support sets prepared for shipment.

4.6.1.2 Sampling. Sampling shall be in accordance with MIL-STD-105. The lot shall contain only one size of packed support sets.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112. Shipping base not designed as specified.
113. Quantity of support sets per pack not as specified for each size.
114. Support sets not arranged in pack as specified.
115. Support sets not secured in pack as specified.
116. Marking incorrect, illegible, or missing.

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5. PACKAGING

5.1 Packing. Packing shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Size 1 support sets. Twelve size 1 support sets shall be packed as shown in figure 2 on a shipping base as shown in figure 1. Twenty-four side section weldments shall be placed upright on the shipping base. Two lateral crossbar weldments shall be positioned on the outer two side section weldments. A spacer, part number 5, shall be inserted between the two center side section weldments to eliminate movement of the weldments. Two pieces of lumber, part number 6, shall be placed over the tubing crossbars of the side section weldments and fastened to the spacer with three 4d nails at each intersection. The steel strapping shall encircle each part number 6 as well as the tubing crossbars and the spacer. Twenty-two lateral crossbar weldments and 12 tied rod weldments shall be placed in neat layers between the angle and tubular crossbars of the upright side section weldments. Part number 2 shall be nailed to the plywood panel with 4d nails spaced approximately 5 inches apart. Part number 4 shall fit flat, cover the top layer of stacked support set components, and shall be nailed to the plywood panel with 6d nails spaced approximately 5 inches apart. The plywood panels with attached lumber parts shall be secured at each end of the pack with steel strapping only, as shown in figure 2.

5.1.1.2 Size 2 support sets. Fifteen size 2 support sets shall be packed as specified in 5.1.1.1 and figures 1 and 2. Thirty side section weldments shall be placed upright on the shipping base.

5.1.1.3 Size 3 support sets. Ten size 3 support sets shall be packed as specified in 5.1.1.1 and figures 1 and 2. Twenty side section weldments shall be placed upright on the shipping base.

5.1.2 Level C. Level C shall be in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Level A. Marking for level A packing shall be in accordance with MIL-STD-129 and shall be placed on the plywood panel at either end of the pack.

5.2.2 Level C. Marking for level C packing shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The support sets covered by this specification are intended to be used with flat wood pallets. The support set assures rigid support over a pallet load of material, prevents crushing and provides for stable, vertical stacking. Size 1 support set is used on standard two-way entry flush type or four-way entry wing type, 40- by 48-inch pallet. When wing type

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pallets are used, a filler should be added to the wing areas between the upper and lower deck boards. Size 2 support set is used on standard two-way entry flush type, 48- by 60-inch pallet. Size 3 support set is used on standard two-way entry flush type, 32- by 40-inch pallet.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Time frame required for submission of the first article (see 3.2).
e. Level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the items should be an initial production. The first article should consist of 4 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

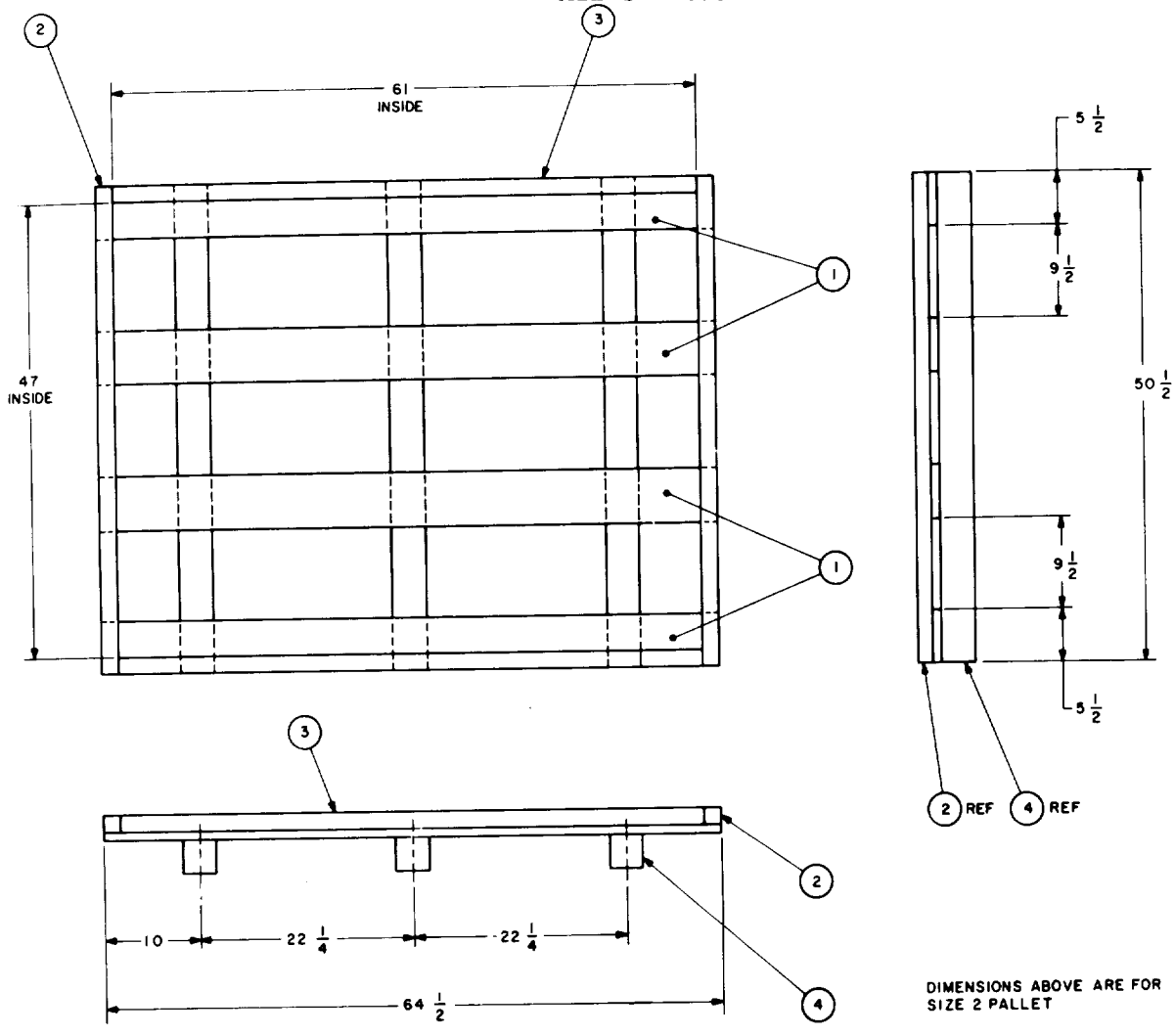
Custodians:
Army - ME
Air Force - 84

Review activity:
Army - SM

Preparing activity:
Army - ME

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DIMENSION TABLE (INCHES)									
FIND NO.	SIZE 1			SIZE 2			SIZE 3		
	LENGTH	WIDTH	THK	LENGTH	WIDTH	THK	LENGTH	WIDTH	THK
1	52 1/2	6	1	64 1/2	6	1	44 1/2	6	1
2	42 1/2	2	2	50 1/2	2	2	34 1/2	2	2
3	49	2	2	61	2	2	41	2	2
4	42 1/2	4	4	50 1/2	4	4	34 1/2	4	4

FIND NO.	QTY	DESCRIPTION
4	3	SKID, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING
3	2	HEADER, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING
2	2	HEADER, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING
1	4	DECKBOARD, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING

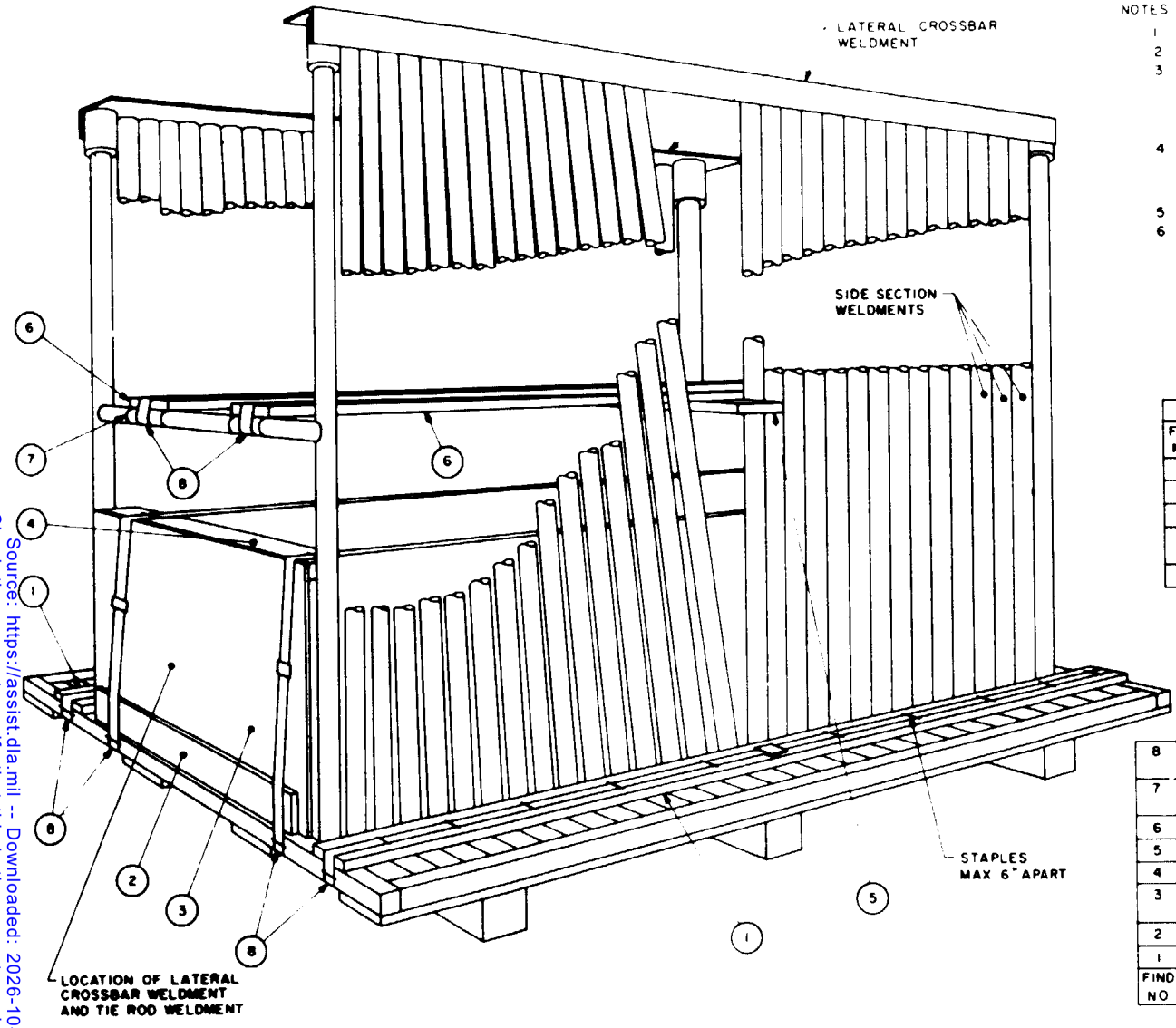
NOTES:

1. SPACING OF PIECES NO. 1 SHALL BE:
   SIZE 1 – 7" (APPROX)
   SIZE 2 – AS SHOWN
   SIZE 3 – 9"
2. FOR SET SIZES 1 AND 3, SPACING OF PIECES NO. 4 SHALL BE ONE AT THE CENTER, AND ONE LOCATED IN FROM EACH END A DISTANCE EQUAL TO APPROX ONE SIXTH THE LENGTH OF THE BASE.
3. LUMBER WIDTHS AND THICKNESSES SHALL BE NOMINAL. LENGTHS SHALL BE ACTUAL.
4. TOLERANCES SHALL BE ±1/8" FOR LENGTH MEASUREMENTS OF PARTS.
5. NAILING OF BASE TO BE AS FOLLOWS:
   FIND NO. 1 AND 4 SECURED TOGETHER AT EACH INTERSECTION WITH FOUR 10d NAILS. PARTS NO. 2 AND 3 SECURED TO NO. 1 WITH TWO 7d NAILS AT EACH INTERSECTION. NAILS TO CONFORM TO FF-N-105, TYPE II, STYLE 7.

FIGURE 1. SHIPPING BASE, LEVEL A, FOR SUPPORT SET, STACKING, PALLET, SIZES 1, 2, AND 3

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NOTES

1. ALL LUMBER SIZES SHALL BE NOMINAL LENGTHS SHALL BE ACTUAL.
2. WIDTH OF PIECE NO 5 SHALL BE SUFFICIENT TO ASSURE A TIGHT FIT.
3. PLYWOOD DIMENSIONS SHALL VARY WITH THE SIZE OF SUPPORT SET AS FOLLOWS
   SIZE 1 - 35" X HEIGHT OF STACK + 1 1/4"
   SIZE 2 - 29 1/2" X HEIGHT OF STACK + 2"
   SIZE 3 - 13 1/2" X HEIGHT OF STACK + 2"
4. PACK OF SIZE 1 SETS SHALL HAVE PIECES NO 1 PLACED INSIDE VERTICAL MEMBERS OF SIDE SECTION WELDMENTS PIECES NO 2 SHALL FIT BETWEEN PIECES NO 1; PLYWOOD SHALL REST ATOP ENDS OF PIECES NO 1.
5. NAILS TO CONFORM WITH FF-N-105, TYPE II, STYLE 7
6. ALL STRAPPING TO BE STAPLED TO WOOD MEMBERS

DIMENSION TABLE (INCHES)

FIND NO.	SIZE 1			SIZE 2			SIZE 3		
	LENGTH	WIDTH	THK	LENGTH	WIDTH	THK	LENGTH	WIDTH	THK
1	51 1/2	4	1	63 1/2	4	1	43 1/2	4	1
2	26	4	1	29 1/2	4	1	13 1/2	4	1
4	32 1/2	3	2	27	3	2	11	3	2
5	37 1/2	SEE NOTE 2	1	33 1/4	SEE NOTE 2	1	17	SEE NOTE 2	1
6	48	4	1	60	4	1	40	4	1

FIND NO	QTY	DESCRIPTION
8	6	STRAPPING, STEEL, FLAT, QQ-S-781, TYPE I, FINISH A, 0.020" X 3/4" MIN X LENGTH AS REQUIRED.
7	4	FIBERBOARD, PPP-F-320, TYPE CF OR SF, CLASS WEATHER RESISTANT
6	2	BATTEN, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING
5	1	SPACER, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING
4	2	CLEAT, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING
3	2	PLYWOOD, NN-P-530, GROUP B (PS 1, GRADE C-D WITH EXTERIOR GLUE); 1/2" THICK (SEE NOTE 3)
2	2	CLEAT, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING
1	2	BATTEN, LUMBER, MIL-STD-731, CLASS 2, MULTIPLE HANDLING

FIGURE 2. Packing, level a, for support set, stacking, pallet, sizes 1, 2, and 3.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-21859E	Support Sets, Stacking, Material Handling Pallet

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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